* election and proton সমান হলে তাকে change neutral বলে।

* e^- বের করে দিলে positive change

  e^- গ্রহন করলে negative change

* কোনো object এ e^- এর সংখ্যা বেড়ে গেলে বা কমে গেলে যেখানে change genarate হয়। এবং যদি change টা move করতে না পারে, এক জায়গায় স্থির থাকে তখন তাকে static electicity.

  আর যদি change গুলো move করতে পারে তখন বলে current flow.

Repulsion

$\left.\begin{matrix} + \; + \\ - \; - \end{matrix}\right\}$ similar type chanege ~~B~~repels each other ~~ঠেলে~~

Attration

$\left.\begin{matrix} + \; - \\ - \; + \end{matrix}\right\}$ They attract each other

unite vector

$$\boxed{F = k\frac{q_1 q_2}{r^2}}$$

→ magnitude of F, with out direction

q_1 ———— q_2

r

$$\boxed{\vec{F} = k\frac{q_1 q_2}{r^2}\hat{r}}$$

K= kulomb constant

$\boxed{K = 8.99\times10^9\ Nm^2/C^2}$ along x-axis/

$\boxed{K = \frac{1}{4\pi\epsilon_0}}$ $\boxed{\epsilon_0 = 8.85\times10^{-12}}$

If the distance between the two changes are doubled what is the change in the force between them?

Solve: $F_1 = k\dfrac{q_1 q_2}{r^2}$ $\quad r_2 = 2r$

$$F_2 = k\frac{q_1 q_2}{r_2^2}$$

$$= k\frac{q_1 q_2}{(2r)^2}$$

$$F_2 = \frac{F_1}{4}$$

Problem: 1

A positive change of 6×10^{-6} C is 0.040m from the second positive change of 4×10^{-6} C. Calculate the force between the changes.

Solve: $\vec{F} = k\dfrac{q_1 q_2}{r^2}\hat{r}$

$q_1 = 6\times10^{-6}$ C

$q_2 = 4\times10^{-6}$ C

$r = 0.04$ m

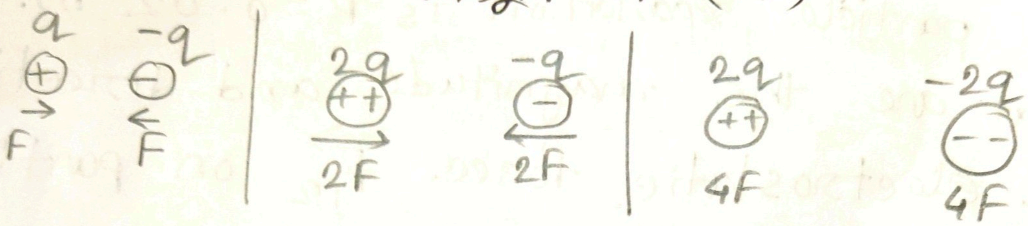
$$= 9\times10^{9}\frac{6\times10^{-6}\times4\times10^{-6}}{(0.04)^2}$$

$$= 135 \text{ N}$$

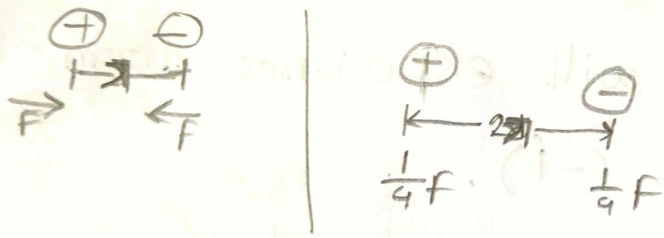
(Ans:)

* 2 টা charge (q^+, q^-) এর মধ্যে একটা charge দ্বিগুণ করলে force দ্বিগুণ হয় (2F).

* 2 টা charge (q^+, q^-) এর দুইটা charge দ্বিগুণ করলে force চারগুণ হয় (4F)



* Distance (r) দ্বিগুণ হলে (2r) force ($\frac{1}{4}F$) হবে।

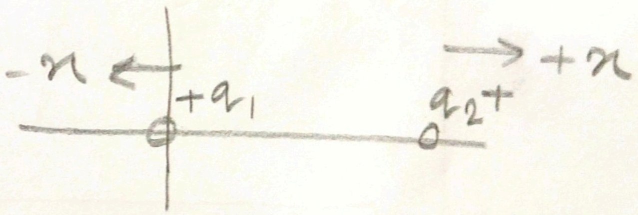


$$\boxed{\begin{aligned} 1P &= 1.6\times10^{-19}\,C \\ 1e^- &= -1.6\times10^{-19}\,C \end{aligned}}$$

* $q_1 > q_2$; $\vec{F} = (+)$ x-axis, দিকে towards q_2
  $q_2 > q_1$; $\vec{F} = (-)$ x-axis, towards q_1

Two positively charged particles fixed in place on an x axis. The charges are $q_1 = 1.60 \times 10^{-19}$ C and $q_2 = 3.2 \times 10^{-19}$ C and the particle separation is $R = 0.02$ m. What are the magnitude and direction of the electrostatic force $\vec{F}_{12}$ on particle 1 from particle 2?

Solve: The q_1 will experience ~~force~~ force along $-x$ axis $(-\hat{i})$.



$$\vec{F}_{12} = k \frac{q_1 q_2}{r^2} \hat{i}$$

$$= 9 \times 10^{9} \frac{(1.6 \times 10^{-19} \times 3.20 \times 10^{-19})}{(0.02)^2}$$

$$= 1.152 \times 10^{-24} \hat{i}$$

Thus the q_1 will experience force along with $-x$ axis, so, we can write ~~$F_{12} = (1.152 \times$~~

$$\vec{F} = -(1.152 \times 10^{-24}) \hat{i}$$

4. The nucleus in an iron atom has a radius of about 4.0×10^{-15} m and contains 26 protons.

(a) What is the magnitude of the repulsive electrostatic force between two of the protons that are separated by 4.0×10^{-15} m?

(b) What is the magnitude of the gravitational force between those same two protons?

Solve

(a) $F = k\frac{e^2}{r^2}$

$e = 1.602 \times 10^{-19}$ C

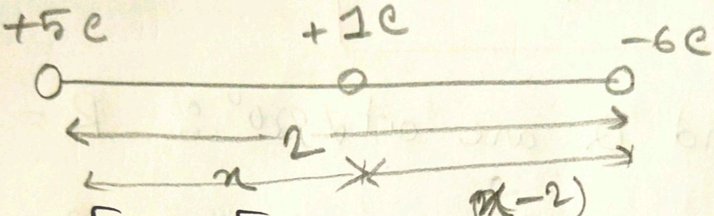
$r = 4 \times 10^{-15}$ m

$$= 9 \times 10^9 \frac{(1.602 \times 10^{-19})^2}{(4.0 \times 10^{-15})^2} = 14.4 \text{ N}$$

(b)

* Same direction হলে $F_{net} = F_1 + F_2$

Different " " $F_{net} = F_1 - F_2$



$F_{net} = F_1 - F_2$

$\Rightarrow \frac{K.5}{x^2} + \frac{K.6}{(x-2)^2} = 0$

$\Rightarrow \frac{5}{x^2} + \frac{6}{(x+2)^2} = 0$

$\Rightarrow 5(x-2)^2 + 6x^2 = 0$

$\Rightarrow 5(x^2 - 4x + 4) + 6x^2 = 0$

$\Rightarrow 11x^2 - 20x + 20 = 0$

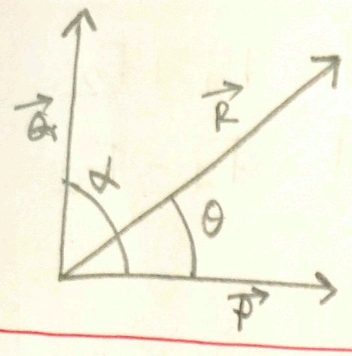
There is no place where....

$F_1 = k\frac{5\times1}{x^2}$

$= \frac{5k}{x^2}$

$F_2 = \frac{k(-6)\times1}{(x-2)^2}$

$= \frac{-6k}{(x-2)^2}$



$$\boxed{\tan\theta = \frac{Q\sin\alpha}{P + Q\cos\alpha}}$$

$$\boxed{R = \sqrt{P^2 + Q^2 + 2PQ\cos\alpha}}$$

$$\boxed{F_{net} = \sqrt{F_1^2 + F_2^2}}$$

when, P and Q are at $\alpha = 90°$: $R = \sqrt{P^2 + Q^2}$

" " " " $\alpha = 0°$: $R = P + Q$

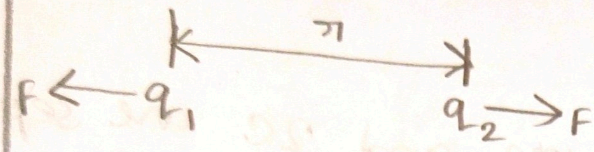
" " " " $\alpha = 180°$: $R = P \sim Q$

[বড়টা থেকে ছোট বিয়োগ]

(youtube)
(My self)

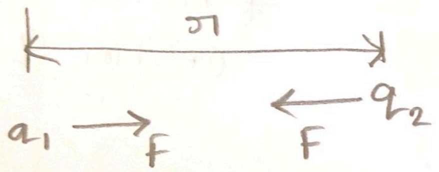
$$F = \frac{1}{4\pi\epsilon_0}\frac{q_1 q_2}{r^2}$$

Case-I: $(q_1 q_2 > 0)$ both change are same polarity
→ repulsive Force

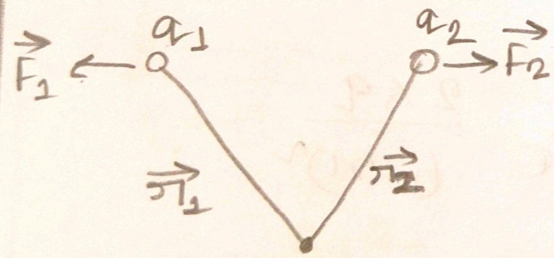


Case-II: $(q_1 q_2 < 0)$ one change positive and other chane negative.

→ attractive force



For Repulsive Force:



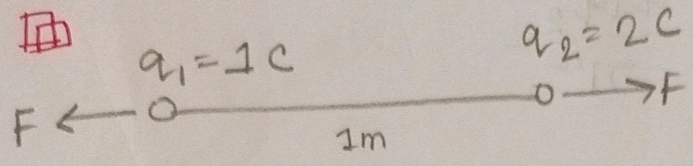
$$F_1 = \frac{1}{4\pi\epsilon_0}\frac{q_1 q_2}{r^2}\hat{r}_1 = \frac{1}{4\pi\epsilon_0}\frac{q_1 q_2}{r^3}\vec{r}$$

$$\hat{r}_1 = \frac{\vec{r}}{|\vec{r}|} = \frac{\vec{r}_1 - \vec{r}_2}{r}$$

$$F_2 = \frac{1}{4\pi\epsilon_0}\frac{q_1 q_2}{r^2}\hat{r}_2 = \frac{1}{4\pi\epsilon_0}\frac{q_1 q_2}{r^3}\vec{r}$$

$$\hat{r}_2 = \frac{\vec{r}}{|\vec{r}|} = \frac{\vec{r}_2 - \vec{r}_1}{r}$$

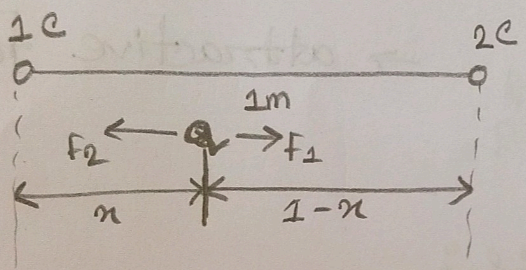
□ What is the force which is active on q1?



Solve:

$$F = \frac{1}{4\pi\epsilon_0}\frac{q_1 q_2}{r^2} = 9\times10^9 \frac{1\times2}{1^2} = 18\times10^9\ N$$

□ If two charges 1C and 2C are separated by 1m, then at what distance from 1C charge, force on charge q will be zero?



* Force সমান হলে সেখানে q = 0 হবে।

$F_1 = F_2$

$$\frac{1}{4\pi\epsilon_0}\cdot\frac{1\times q}{x^2} = \frac{1}{4\pi\epsilon_0}\frac{2\times q}{(1-x)^2}$$

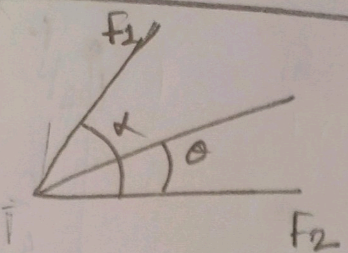
$\Rightarrow q(1-x)^2 = 2qx^2$

$\Rightarrow 1^2 - 2x + x^2 = 2x^2$

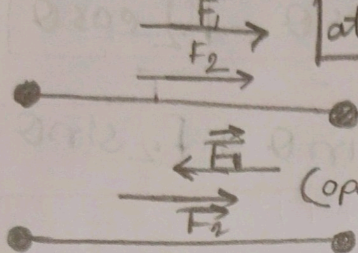
$\Rightarrow x^2 + 2x - 1 = 0$

$x = (0.41, -2.41)$

$\therefore$ 0.41 distance q will be zero.



$$F_{net} = \sqrt{F_1^2 + F_2^2}$$



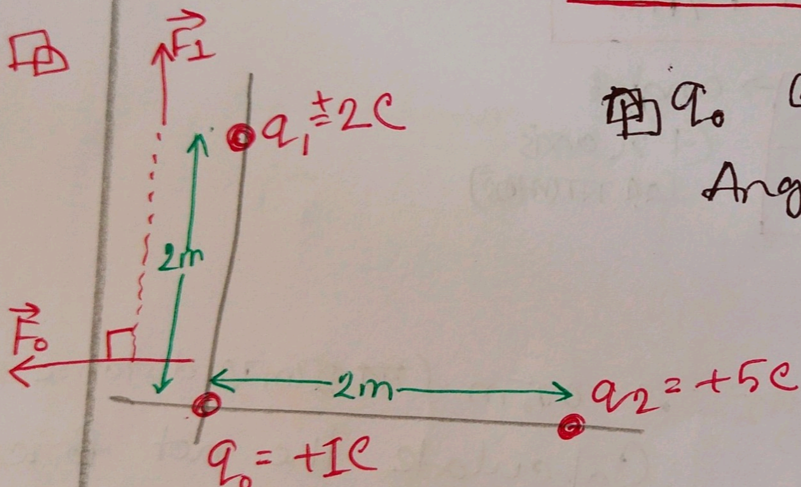
(same direction)

$\boxed{\text{at } 0^\circ \text{ or } 180^\circ}$

$\boxed{F = F_1 + F_2}$

(oposite direction)

$\boxed{F = F_1 \sim F_2}$



q_0 এর উপর Force net কত?
Angle?

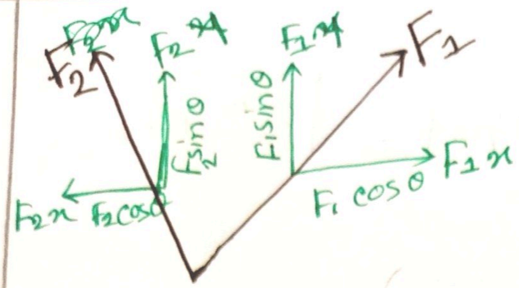
$$F_1 = K\frac{q_1 q_0}{2^2} = 9\times10^9 \frac{2\times1}{2^2} = 4.5\times10^9 \text{ N}$$

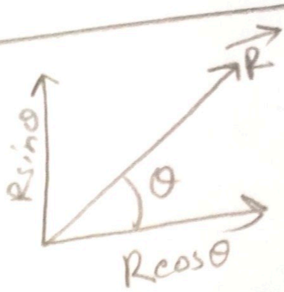
$$F_2 = K\frac{q_2 q_0}{2^2} = 9\times10^9 \frac{5\times1}{2^2} = 1.125\times10^{10} \text{ N}$$

$$F_{net} = \sqrt{F_1^2 + F_2^2} = \sqrt{(4.5\times10^9)^2 + (1.125\times10^{10})^2}$$
$$= 1.21\times10^{10} \text{ N}$$

$\alpha = 90^\circ$,

$$\theta = \tan^{-1}\frac{F_2 \sin\alpha}{F_1 + F_2\cos\alpha} = \tan^{-1}\frac{1.125\times10^{10}\sin 90^\circ}{4.5\times10^9 + 1.125\times10^{10}\cos 90^\circ}$$
$$= 68.199^\circ$$



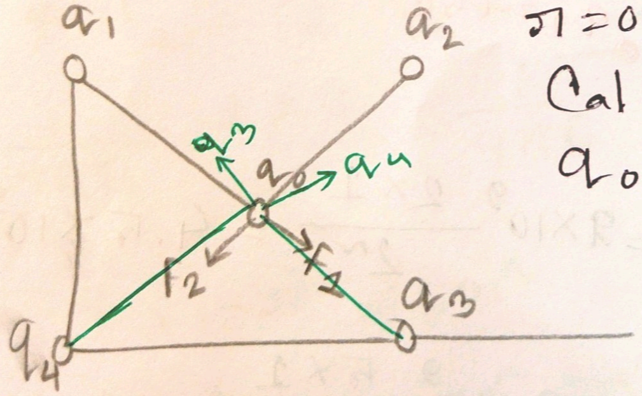


$$\boxed{F_xNet = F_1x - F_2x = F_1\cos\theta - F_2\cos\theta}$$

$$\boxed{F_yNet = F_1y + F_1y = F_1\sin\theta + F_2\sin\theta}$$

$$\boxed{F_{Net} = \sqrt{F_xNet^2 + F_yNet^2}}$$

$$\boxed{\tan\theta = \frac{F_yNet}{F_xNet}}$$ → angle (+x axis এর সাপেক্ষে)



$r = 0.5m$ (সব গুলোর distance same)

Calculate the net force o

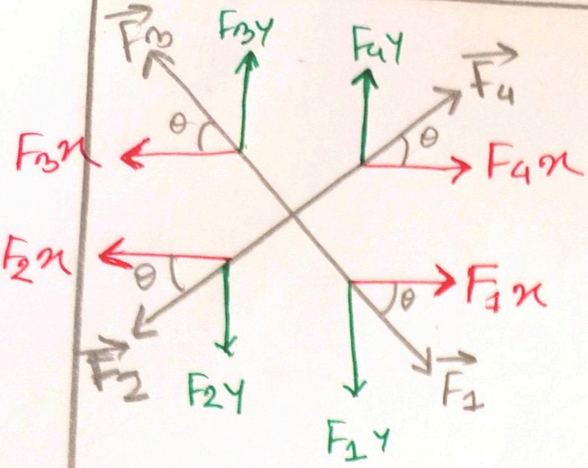
q_0?

$$F_1 = k\frac{q_1q_0}{r^2}$$

$$F_2 = k\frac{q_2q_0}{r^2}$$

$$F_3 = k\frac{q_3q_0}{r^2}$$

$$F_4 = k\frac{q_4q_0}{r^2}$$



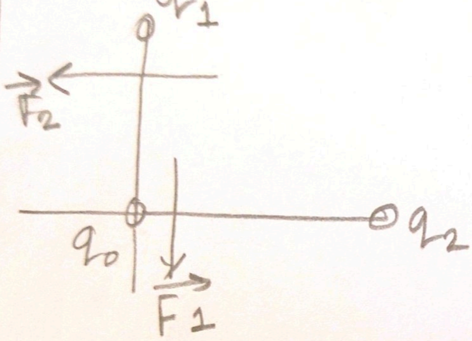
$$F_x Net = F_1x + F_4x - F_2x - F_3x$$

$$F_y Net = F_3y + F_4y - F_1y - F_2y$$

$$F Net = \sqrt{F_x Net^2 + F_y Net^2}$$

$$\tan\theta = \frac{F_y Net}{F_x Net}$$

Case : I

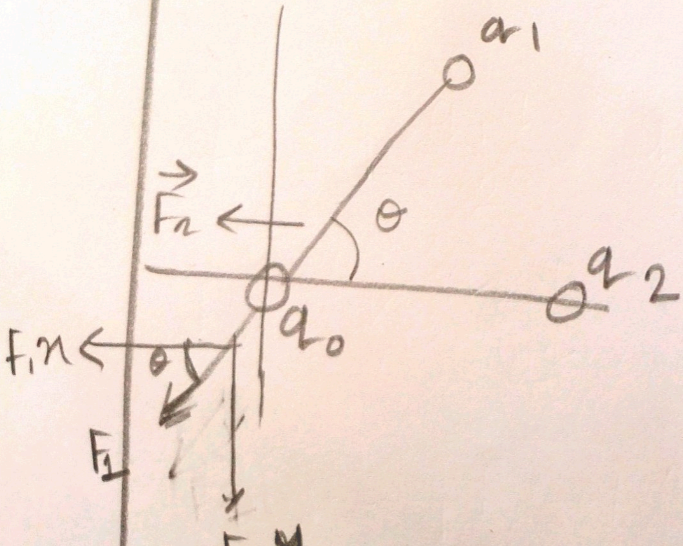


$$F_x Net = F_1x - F_2x$$
$$= -F_2x$$
$$= -F_2$$

$$F_y Net = -F_1y + F_2y$$
$$= -F_1y = -F_1$$

$$F Net = \sqrt{F_x Net^2 + F_y Net^2}$$

Case : II



$$F_x Net = -F_1x - F_2x$$
$$= -F_1\cos\theta - F_2$$

$$F_y Net = F_1y = F_1\sin\theta$$

$$F Net = \sqrt{F_x Net^2 + F_y Net^2}$$